# Tetrachloroethylene (PCE, PERC) & Trichloroethylene (TCE): An Overview

Tetrachloroethylene (PCE) and Trichloroethylene (TCE) are chlorinated liquid hydrocarbons, primarily used as degreaser agents for the cold cleaning of fabricated metal parts. They serve as general solvents for resins, fats, oils, waxes, rubber, paints, cellulose esters and ethers, and varnishes [1]. In the industry, both chemicals are of occupational concern because of their high toxicity levels, and widespread usage; they have recently become the primary pollutants in most of the state and federal Superfund sites [2]. Moreover, traces of these harmful compounds have been found to occur in underground and surface water sources as a result of inappropriate manufacturing, consumption, and disposal practices [3].

Chlorinated solvents are usually difficult to biodegrade, and they pose a significant and continuing threat to the United States' potable water supplies, as well as the worker's welfare. Both PCE and TCE have been associated with toxic effects in the kidney and liver, and can also cause depression in the central nervous system, reproductive dysfunction, higher cancer incidence, among other diseases. In this regard, business owners and consumers, who are still employing these compounds, should keep an eye on the latest legislation and potential implications when using the chemicals, as well as other alternatives they could have.

## TETRACHLOROETHYLENE (PERCHLOROETHYLENE)

### What is Tetrachloroethylene?

Tetrachloroethylene, also known as perchloroethylene (PCE or PERC), is a nonflammable colorless liquid used for dry cleaning operations and as a starting material to synthesize other compounds. It has been associated with mild side effects after short-term exposure such as upper respiratory irritation, reversible mood, dizziness, sleepiness, headaches, and coordination impairment. Likewise, long-term exposure leads to severe adverse reactions such as cognitive damage, kidney & liver diseases, immunotoxicity, reproductive issues, and potential cancer development. As a result, the US Environmental Protection Agency (EPA) and other international organizations have classified it as *likely to be carcinogenic to humans* [4].

In the United States, PCE production peaked at 350 million kg in 1980 when it was introduced as a replacement alternative to trichloroethylene (TCE) for metal cleaning, vapor degreasing, textile processing, dry cleaning, and as a chemical precursor [5]. Since then, tetrachloroethylene production has been declining because of its toxic and rapid evaporation attributes, which have made it detectable in ground and surface water sources, air, soil, food, and even breast milk samples. The latest industrial reports indicate that 215 manufacturing companies have been continuously producing close to 65 million lbs. of PCE annually, and 22 other facilities import 26.5 million lbs. of the chemical into the country [6].

Recently, Superfund site investigations have been conducted involving the nearby communities to take robust response actions against tetrachloroethylene contamination. The first water source detection occurred in the 1990s during a routine sampling practice, and the number has been increasing since then with evidence of this substance in nearly 28% of the sites today. The major concerns resulting from inappropriate disposal of

hazardous chemicals are the long-term cleanup efforts, leakage to other groundwater sources, and health risks for both humans and animals [7].

## Tetrachloroethylene (PCE) Toxicity Case Studies

Multiple case studies about the side effects of tetrachloroethylene (PCE) have been reported throughout the years due to its prevalence in the market. The toxic levels and exposure pathways may change depending on the type of clinical trial that was conducted or public release addressing major contamination evidence. Table 1 presents, in chronological order, the most relevant incidents related to tetrachloroethylene exposure, including the conditions in which they occurred, as well as the average toxic levels for each case.

| **ACUTE TOXICITY & IRRITATION** | | | | |
|---|---|---|---|---|
| **Case (Year)** | **N° of Patients** | **Symptoms** | **Job Description** | **Toxic Levels** |
| 1985 [8] | 1 child | Somnolence and hyperventilation | Accidental ingestion | 30 µg in the blood |
| 1990 [9] | 22 males | The visual function was mildly affected by Per exposure, which interfered with the central nerve conduction | Medical experiment volunteers | 10 and 50 ppm, during 4 hours for 4 days. |
| **NEUROTOXICITY** | | | | |
| **Case (Year)** | **N° of Patients** | **Symptoms** | **Job Description** | **Toxic Levels** |
| 2011 [10] | 1512 people | Risks of autism spectrum disorders, and development of drug and alcohol consumption | Residents of Cape Cod, Massachusetts towns | Water samples with PCE up to 80 ug/L |
| 2012 [11] | 99 pairs of twins | Parkinson's disease symptoms | World War II Veteran Twins | No info available |
| 2012 [12] | 619 participants | Early childhood exposure was found to be associated with long-term visual decrements in adulthood | Residents of Cape Cod, Massachusetts towns | Water samples with PCE up to 80 ug/L |
| 2014 [13] | > 150.000 people from 1975 - 1985 | Workers exposed to polluted water were more likely to further develop different cancer types | Marine and Naval personnel at USMC base Camp Lejeune | 215 µg/L in drinking water |
| 2015 [14] | 50 Employees | Depression of the central nervous system; including dizziness and drowsiness. | Dry-Cleaning | Atmospheric PCE of 7 ppm |
| **KIDNEY TOXICITY** | | | | |
| **Case (Year)** | **N° of Patients** | **Symptoms** | **Job Description** | **Toxic Levels** |
| 1983 [15] | 438 workers | Mild renal tubular lesions rather than glomerular | Dry cleaning shops | 10 ppm of PERC |
| 1999 [16] | 82 workers | Minor renal tubular damage | Dry cleaning | 2.2 - 44.6 mg/m$^3$ PERC |
| 2000 [17] | 40 females | PERC induced dose-degrading effects on the kidneys | Ironing shop and dry cleaning | 60 – 240 mg/m$^3$ PERC |
| **LIVER TOXICITY** | | | | |
| **Case (Year)** | **N° of Patients** | **Symptoms** | **Job Description** | **Toxic Levels** |

| | | | | |
|---|---|---|---|---|
| 1992 [18] | 141 workers | Occupational exposure might cause early hepato-biliary changes | Small laundries and dry cleaning shops | PCE levels of 50 ppm on average |
| 1995 [19] | 5 workers | Mild to moderate hepatic parenchymal changes | Dry Cleaning | PCE up to 83 ppm |
| **IMMUNE AND HEMATOLOGICAL TOXICITY** | | | | |
| **Case (Year)** | **N° of Patients** | **Symptoms** | **Job Description** | **Toxic Levels** |
| 1996 [20] | 279 patients | Connective tissue diseases, systemic sclerosis, and rheumatoid arthritis | Dry cleaning and aircraft industry workers | 3899 mg/m3 |
| 2003 [21] | 660 female workers | Scleroderma-like symptoms | Dry cleaning and aircraft industry | Not specified |
| 2010 [22] | 80 adult males | Immunotoxicity leads to allergic diseases and autoimmune reactions | Dry cleaning | PCE up to 265 $\mu g/m^3$ |
| 2015 [23] | 175 cases | Primary Sjogren's syndrome (PSS) symptoms | Multiple jobs, with common exposure | Not specified |
| **REPRODUCTIVE & DEVELOPMENTAL TOXICITY** | | | | |
| **Case (Year)** | **N° of Patients** | **Symptoms** | **Job Description** | **Toxic Levels** |
| 1990 [24] | 1117 female patients | Low birth weight, congenital malformations, and spontaneous abortions | Laundry and dry-cleaning workers | Doses of 2 – 20 hours per week |
| 1991 [25] | 1926 pregnant female patients | They had headaches, dizziness, and forgetfulness. Miscarriage risk was double, and triple for congenital anomalies | Household use of solvent-containing products | Doses of 10 hours per week |
| 1991 [26] | 34 workers | Subtle effects on sperm quality, with less linearity in sperms swimming paths | Dry cleaning | Breathing samples of 2.67 µg/m3 |
| 2009 [27] | 1658 children | risk of congenital anomalies, such as eye, ear irregularities, and oral cleft defects | Households with contaminated water supplies | Water samples with PCE up to 80 µg/L |
| **CARCINOGENICITY STUDIES** | | | | |
| **Case (Year)** | **N° of Patients** | **Symptoms** | **Job Description** | **Toxic Levels** |
| 2011 [28] | 1704 workers | Exposure to PCE was associated with an increased risk of lung, esophagus, and tongue cancer | Dry workers in 4 US cities (San Francisco, Chicago, Detroit, and New York) | More than 1 year of occupational exposure |
| 2013 [29] | Children born during 1968 - 1985 | Associations between drinking water contaminants and childhood hematopoietic cancers, and neural tube defects (NTDs) | Marine Corps Base Camp Lejeune personnel | Maximum peaks of 215 ppm PCE |
| 2014 [30] | > 150.000 people from 1975 - 1985 | Workers exposed to polluted water were more likely to further develop different cancer types | Marine and Naval personnel at USMC base Camp Lejeune | 215 µg/L in drinking water |
| 2016 [31] | 3851 incidents 1999 - 2008 | Residents close to sites were at higher risk of having Diffuse large B cell lymphoma (DLBCL) | Residents from a Georgia town | Not specified |

| | | | | |
|---|---|---|---|---|
| 2017 [32] | 775 female workers | Exposure to PCE may increase the risk of head and neck squamous cell carcinomas in women | Electrical assemblers, printers, welders, cutters, and equipment manufacturing | At least 1 month of continuous exposure |

**Table 1.** Case Studies on Tetrachloroethylene (PCE)

## TRICHLOROETHYLENE

### What is trichloroethylene?

Trichloroethylene (TCE) is a volatile organic solvent used in multiple manufacturing industries (e.g., aircraft, spacecraft, electronic) to remove grease from metal parts. This colorless liquid has been associated with negative health effects such as upper respiratory irritation, kidney & liver dysfunction, developmental defects, immunosuppression, neurobehavioral changes, cancer development, among many others. Although TCE can be highly convenient for vapor degreasing of metal parts, governmental entities have listed it as a hazardous chemical substance that could leach into water supplies, feed, and air [33].

Along with Perchloroethylene, Trichloroethylene has also been widely produced and commercialized since the 1920s. On a global scale, roughly 80-90% of TCE production goes exclusively to the degreasing industry, being used in paints, adhesives, varnishes, and lacquer formulations. In the United States, TCE production peaked at 280 million kg in 1970 when it had been used in the food industry for decaffeination; in cosmetics as an extractant; in pesticides for spotting fluids; and even in pharmaceutics as a volatile anesthetic. Industrial working settings are the most usual way to get exposed to trichloroethylene, which has led to serious occupational health concerns [34].

Environmental fate testing shows that TCE can easily penetrate surface soils through volatilization, which makes it likely to migrate into groundwater sources. As a result of its prevalence in the environment, the general population − especially workers − can get exposed via ingestion, inhalation, and contact. Up to 2011, this toxic compound has been identified in more than 760 Superfund sites, and the ATSDR reported that between 9 to 34% of the drinking water supply sources in the country exhibited some TCE contamination [35]. Moreover, the National Health and Nutrition Examination Survey suggests that nearly 10% of Americans have detectable levels of TCE in their blood [36].

### Trichloroethylene (TCE) Toxicity Case Studies

The Environmental Protection Agency (EPA) has consistently released studies about the hazards that TCE can impose on humans, animals, and the environment. These reviews classify all health hazards in different categories according to the human tissues that are being or could be affected. Table 2 presents, in chronological order, the most relevant incidents related to trichloroethylene exposure, including the conditions in which they occurred, as well as the average levels of the compound for each case.

| NEUROTOXICITY |
|---|

| Case (Year) | N° of Patients | Symptoms | Job Description | Toxic Levels |
|---|---|---|---|---|
| 1982 [37] | 11 workers | Symptoms of trigeminal nerve impairment | Industrial workers | Not specified |
| 1990 [38] | 2 patients | Mild to moderate cognitive, and psychomotor impairments | Mixing metals in an electronics company | High but not specified |
| 1999 [39] | 4041 people | Links between TCE and speech, and hearing impairment | Contaminated drinking water | 50 – 500 ppb |
| 2002 [40] | 236 residents | Neurobehavioral impairments, an elevated profile of mood state scores, and excessive symptom frequencies | Living close to electronic manufacturing plants | 1 – 100 ppm |
| 2003 [41] | 143 residents | Long-term exposure to TCE is associated with neurobehavioral deficits | Contaminated municipal water supply | TCE >15 ppb |
| **KIDNEY TOXICITY** | | | | |
| **Case (Year)** | **N° of Patients** | **Symptoms** | **Job Description** | **Toxic Levels** |
| 1993 [42] | 29 workers | Slight and severe tubular damage in the kidneys | Metal manufacturing | TCE < 50 mg/m$^3$ |
| 1999 [43] | 39 workers | TRI caused persistent changes to the tubular system of the kidney | Cardboard factory workers | TCE is around 500 ppm |
| 2004 [44] | 70 workers | High risk of kidney damage at concentrations >250 ppm | Hospital and administrative staff | TCE 250 ppm |
| **LIVER TOXICITY** | | | | |
| **Case (Year)** | **N° of Patients** | **Symptoms** | **Job Description** | **Toxic Levels** |
| 2003 [45] | 155 male patients | The link between exposure and male liver cancer symptoms | Exposure to contaminated soil and groundwater | TCE up to 1100 mg/kg |
| 2010 [46] | 1 male patient | Experienced jaundice, fever, red sore eyes, and widespread rashes | Automated degreasing machines | 22 mg/L at work |
| **IMMUNE AND HEMATOLOGICAL TOXICITY** | | | | |
| **Case (Year)** | **N° of Patients** | **Symptoms** | **Job Description** | **Toxic Levels** |
| 2003 [47] | 347 Danish companies | High risk to develop symptoms of non-Hodgkin's lymphoma | Iron and metal dry cleaning | 75 – 318 mg/m$^3$ in air |
| 2008 [48] | About 14500 workers | Positive associations with several types of cancer (non-Hodgkin's lymphoma) | Aircraft maintenance workers | Not specified |
| 2012 [49] | 80 people exposed | A decline in peripheral blood cell counts (lymphocytes, B cells, and CD4+ T cells) | Metal degreasing and contaminated groundwater | Both low TCE <12ppm and high ≥12ppm |
| **REPRODUCTIVE & DEVELOPMENTAL TOXICITY** | | | | |
| **Case (Year)** | **N° of Patients** | **Symptoms** | **Job Description** | **Toxic Levels** |
| 1973 [50] | 184 women | Increase in amenorrhea | Assembling small electrical parts | Mean TCE of 200 mg/m$^3$ |
| 1978 [51] | A 42-year-old man | Impotence and gynecomastia | Aircraft mechanic | Not reported |
| 1995 [52] | 197 women | Reduced incidence of fecundability in high exposure | Industrial hygienists | High but not specified |

| | | | | |
|---|---|---|---|---|
| 1996 [53] | 13 male workers | A low percentage of normal sperm morphology | Electronic factory, metal degreasing | Mean TCE of 30 ppm |
| 2001 [54] | 75 men and 71 women | Altered libido, and increase in abnormal menstrual cycle | Contaminated groundwater | Not specified |
| **CARCINOGENICITY STUDIES** | | | | |
| **Case (Year)** | **N° of Patients** | **Symptoms** | **Job Description** | **Toxic Levels** |
| 1995 [55] | 169 men | Over prolonged exposure time may lead to renal tumors | Cardboard factory | Not reported |
| 1995 [56] | 1391 women | Significant increase in cervix uteri, lymphatic, and hematopoietic cancers | Finnish maintenance workers | Not specified |
| 1998 [57] | 14457 workers | Moderate risk of breast cancer | Aircraft maintenance | TCE < 15 ppm |
| 2003 [58] | 40049 workers | Elevated risk for non-Hodgkin's lymphoma in offspring's | Blue-collar worker | TCE of 58 mg/L in urine |
| 2005 [59] | Close to 54500 cases | The solvent was positively associated with breast cancer in women and men | Residents in Texas | Not reported |
| 2005 [60] | 6107 male workers | Likely to develop cancer of the lungs, melanoma, esophagus, kidney, and stomach | Aerospace company workers | Not specified |
| 2007 [61] | 276 men | TCE exposure showed a positive relation with prostate cancer | Aerospace and radiation workers | Not specified |
| 2008 [62] | 40647 female workers | Childhood leukemia is related to mothers exposed during pregnancies | Electronics factory | Not specified |

**Table 2.** Case Studies on Trichloroethylene (TCE)

## Safety Regulations for PCE and TCE

As a response to the numerous human studies on PCE and TCE, many organizations have set recommendations addressing their usage. The most relevant national and international guidelines, regulations, and advisories regarding tetrachloroethylene and trichloroethylene are summarized in Table 3. The values represent the threshold levels that employers, importers, recyclers, and sellers should follow to avoid legal sanctions and protect the public and workers from adverse health effects from exposure. For the EPA and DOE guidelines, each consecutive AEGL and PAC number is associated with a progressively severe effect that involves a higher exposure level to the chemicals.

| **Regulation** | | **Tetrachloroethylene** | **Trichloroethylene** |
|---|---|---|---|
| **National** | OSHA | Permissible exposure limit (PEL) of 100 -200 ppm, and a 300 ppm max peak (< 5min each 3h) | Permissible exposure limit (PEL) of 100 -200 ppm, and a 300 ppm max peak (< 5min each 2h) |
| | NIOSH | Exposure should be minimized as much as possible | Recommended exposure limit (REL) of 2 ppm/h or 25 ppm (10h TWA) |

| Regulation | | | Tetrachloroethylene | Trichloroethylene |
|---|---|---|---|---|
| | EPA | Hazardous Pollutant | YES | YES |
| | | AEGL-1 | 35 ppm (8 hours) | 77 ppm (8 hours) |
| | | AEGL-2 | 81 ppm (8 hours) | 240 ppm (8 hours) |
| | | AEGL-3 | 410 ppm (8 hours) | 970 ppm (8 hours) |
| | DOE | PAC 1 | 35 ppm | 130 ppm |
| | | PAC 2 | 230 ppm | 450 ppm |
| | | PAC 3 | 1200 ppm | 3800 ppm |
| | ACGIH | | Confirmed animal carcinogenic with unknown relevance to humans | Suspected human carcinogen |
| **International** | IARC | | Probably carcinogenic to humans | Carcinogenic to humans |
| | WHO | Air | 0.25 mg/m3 | 2.3 µg/m |
| | | Drinking water | 0.04 mg/L | 0.02 mg/L |

**Table 3.** Regulations and guidelines applicable to tetrachloroethylene [63] and trichloroethylene [64]

In the upcoming years, these regulations could significantly alter the production, usage, transport, storage, and disposal of cleaning products that contain chlorinated chemicals. Official statements about maximal levels permitted in food, water, plants, atmosphere, soil, and animal and human tissues are continuously modified and may not reflect the regulatory status for the chemicals in the future. Companies and customers who employ PCE and TCE for their daily operations should pay special attention to the general discontinuation, management efforts, and feasible alternatives in the market.

## Alternatives to 1-Bromopropane Based Degreasers

Tetrachloroethylene and Trichloroethylene-based degreasers have been popular for industrial cleaning applications because they quickly dissolve oils and greases, evaporate quickly, and have a relatively low cost. As regulations on these compounds continue to tighten on the path to a full ban, Chemtronics continues to engineer alternatives with all the advantages of PCE and TCE, but without the harmful health effects.

Chemtronics Electro-Wash Tri-V Precision Cleaner and Max-Kleen Tri-V Head-duty Degreaser are nonflammable cleaners that quickly remove flux, grease, oils, dirt, dust, and other contaminants from electronic components, metal parts, tools, and assemblies. They remove all types of oil and grease while evaporating quickly and leaving no residues. Tri-V nPB replacement chemistry is an innovative chemistry that does not contain n-propyl bromide, TCE, hazardous air pollutants, or ozone-depleting compounds.

For more information, contact your Chemtronics application specialist at 678-928-6534 or askchemtronics@chemtronics.com.

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